Intelligent Network Security & Digital Twin SOC

1. Project Overview

INSOC-DT combines enterprise networking, intrusion detection, SIEM, machine learning, automated incident response, and a network digital twin for simulation and what-if analysis.

2. Problem & Solution

Problem: Manual monitoring delays threat containment. Traditional alert-only systems require administrator action.

Solution: Transform workflow from `Detect → Alert → Manual` to `Monitor → Detect → Analyze → Correlate → Risk Score → Decide → Respond → Verify` .

3. Objectives

- Network monitoring & intrusion detection
- ML-based threat classification
- Risk scoring & incident management
- Automated IP blocking & host isolation
- Digital twin for simulation & what-if analysis
- SOC dashboard with role-based access

4. High-Level Architecture

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Internet → Firewall → Networks → Sensors (Suricata/Zeek)
→ Event Collector → Detection (Rules + ML) → Correlation
→ Risk Engine → Incident/Response → Firewall/Quarantine → Dashboard

Parallel: Network Telemetry → Digital Twin → Simulation → Impact Analysis → Dashbo
```

5. Network Infrastructure

Lab: EVE-NG / GNS3 with virtual routers, switches, firewall, endpoints, and Kali Linux attacker.

VLAN	Name	Subnet
10	Management	10.10.10.0/24
20	Users	10.10.20.0/24
30	Servers	10.10.30.0/24
40	Security	10.10.40.0/24
50	Quarantine	10.10.50.0/24

6. Detection Engine

- **Signature:** Port scanning, brute force, suspicious payloads
- **Behavioral:** Unusual connection rates, traffic spikes, abnormal DNS

7. Machine Learning

Purpose: Second detection layer enhances, not replaces, deterministic rules.

Features: Flow duration, packets/bytes, ports, protocol, connection count, failed connections, DNS count.

Models: Random Forest (baseline), compared with Decision Tree, Logistic Regression, XGBoost.

Metrics: Accuracy, Precision, Recall, F1 Score, Confusion Matrix, False Positive Rate.

8. Risk Scoring & Automated Response

Risk Score = f(severity, confidence, frequency, asset criticality)

Score	Level	Action
0–30	Low	Log only
31–60	Medium	Alert
61–80	High	Alert + temporary block

Score	Level	Action
81–100	Critical	Automated response

Response Actions: IP blocking, firewall rules, host isolation, quarantine VLAN, incident creation.

Safety: Allowlists, thresholds, expirations, rollback, audit logs.

9. Digital Twin

Software model representing network devices, links, IPs, VLANs, routes, ACLs, firewall rules, and dependencies.

What-If Analysis: Simulate router failure, OSPF changes, link disablement, ACL/firewall modifications.

Output: Affected devices/services, alternative paths, warnings, recommendations.

10. Incident Management

Lifecycle: Detection → Analysis → Containment → Eradication → Recovery → Closure

Record: ID, title, attack type, source/target, severity, risk score, evidence, response, status, notes.

11. SOC Dashboard

- Security overview, active alerts, incidents
- Top attacking IPs, attack timeline
- Network health, blocked IPs, isolated hosts
- Sensor status, response history

Roles: Admin, SOC Analyst, Network Engineer.

12. Evaluation Metrics

- **IDS:** Detection Rate, False Positive Rate, Precision, Recall, F1
- **ML:** Accuracy, Precision, Recall, F1, Confusion Matrix, Inference Time

- **Response:** Detection Time, Response Time, Containment Success, Rollback Success
- **Network:** Latency, Packet Loss, Throughput, CPU/Memory
- **Digital Twin:** Model agreement, connectivity validation, prediction accuracy

13. Final Demo Flow

1. Show healthy network
2. Generate controlled port scan
3. Show Suricata/Zeek evidence
4. Show detection, ML classification, correlation, risk score
5. Execute automated blocking
6. Verify attacker containment
7. Open Digital Twin
8. Simulate router failure or routing change
9. Show impact analysis and recommendations
10. Return to SOC dashboard — review incident timeline